Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 40-tooth driver turns a 240-tooth gear. Give ideal output/input torque and speed ratios and state the loss assumption.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: I = 0 with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw your final wiring route. Mark soldered joints, insulation, strain relief and the switch location. Identify one possible wheel or gear interference point.



Marzano check - circle: ① ② ③

04 Apply the physics

A simplified model gives the car a constant 0.30 N net forward force. Compare acceleration at masses 0.20 kg and 0.25 kg. Complete the table, calculate the percent decrease and state why this fixed-force prediction may differ from a real motor test.

Design	Mass (kg)	Net force (N)	Acceleration
Before	___	___	___
After	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A wire rubs a wheel. Explain the possible effect on energy transfer and propose a repair before another powered test.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Near an ideal long straight wire, current becomes 4 times as large and distance becomes 6 times as large. Compare B.
Mark the distances and field directions you compare.

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

07 Exit check

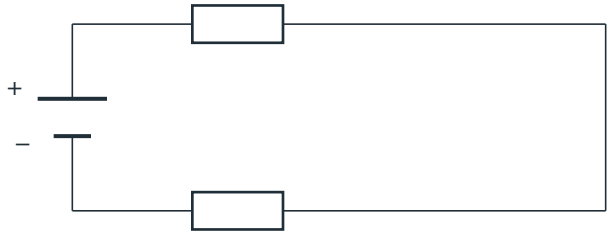
Explain why “lighter is always faster” is too strong for a powered car. Name two other influential factors.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A wire fault adds 2 ohms in series to a supplied motor model of 4 ohms on 6 V. Compare current before/after and power in the unwanted resistance. Explain why a warm joint deserves investigation.



Marzano check - circle: 1 2 3